

Collaborative Localization Using Multiple UAVs: A Deep Denoising Autoencoder-Based Data Compression Approach

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Abstract—In recent years, the deployment of multiple Unmanned Aerial Vehicles (UAVs) for source localization has significantly increased in military, civil, and commercial domains. The transmission of data from the source to each UAV introduces channel noise, which affects localization accuracy. Furthermore, since direct estimation of time of arrival is impractical, signal processing at a central UAV is essential to estimate time difference of arrival to achieve a precise source localization. This processed localization information is subsequently shared among the UAV network. However, transmitting original data to a central point presents challenges due to its time-consuming and low energy efficiency, and highlights the importance of efficient data compression in multiple UAV systems. Consequently, it is recommended that low-dimensional data be transmitted across UAV networks instead of the original high-dimensional signal samples to realize the reconstruction of the original data for source localization. However, as the compression rate (CR) increases, accurately reconstructing the original data becomes challenging. This paper proposes a deep denoising autoencoder (DAE) approach for high-ratio data compression in multiple UAV systems. The DAE not only effectively reduces noise from signals but also significantly decreases the volume of data required for transmission, enabling accurate reconstruction of the original data. Simulations were performed to assess the localization accuracy of the proposed DAE approach. The results indicate that the approach achieves high localization accuracy, particularly in challenging scenarios of low signal-to-noise and high CRs.

Index Terms—Unmanned aerial vehicles (UAVs), collaborative localization, data compression, denoising autoencoder (DAE).

I. INTRODUCTION

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RECENTLY, unmanned aerial vehicles (UAVs) have attracted significant attention due to the rapid development of UAV-related technologies [1], [2]. Precise location sensing is crucial for a wide range of location-based services, given its essential role in these advancements. When multiple UAVs are deployed for source location estimation, they collaborate to perform localization tasks within specified areas of interest. This effective collaboration between multiple UAVs not only mitigates the limited sensing capabilities of individual UAVs but also enhances the fault tolerance of the network, ensuring cost-effective task completion. Due to these beneficial characteristics, multiple UAV collaborative systems are now widely used in both civilian and military areas, supporting tasks such as aerial photography, military reconnaissance, topographic surveys, target tracking, and threat neutralization [3].

The determination of unknown source locations by UAVs has become a significant topic in recent years. Traditional source localization techniques typically follow a two-step approach [4], which are considered indirect processing methods. Initially, these methods employ the generalized cross-correlation method [5] to estimate the time difference of arrival (TDOA). Subsequently, the localization problem is formulated as an optimization problem, constrained by the geometric relationships between the sources and UAVs. These approaches require collaboration among multiple UAVs. However, the practical application of these strategies frequently faces two major challenges. One primary issue is the error in TDOA estimation caused by factors such as multipath propagation, shadowing, and non-line-of-sight (NLOS) propagation, which affects the communication between UAVs and target. This issue is frequently attributed to limited bandwidth and signal strength, which reduces localization accuracy [6]. Additionally, successful collaborative localization relies on efficient data sharing among UAVs. Enhancing the capability to transmit and process all the collected data would significantly improve this process [7]¹. However, communication networks often represents a significant limitation in achieving these goals in real time and with high localization accuracy.

Traditional approaches often fail to address both challenges simultaneously. Recently, deep learning has played an significant role in compressing and reconstructing data. Autoencoders have been shown to provide excellent per-

¹There are some related works considering other aspects. In [8], strategies are proposed to improve communication efficiency in aerial networks. In [9], the deployment of unmanned ground vehicles as mobile charging stations has been suggested to wirelessly support UAVs, ensuring continuous data collection. In [10], a multiple UAV control strategy has been proposed, using deep reinforcement learning to optimize UAV trajectories, addressing mobility control, energy limitations, and communication range issues.

formance with fewer trainable parameters [11]. Their non-linearity enables effective dimensionality reduction, feature extraction, signal compression, and noise reduction [12]–[14]. Furthermore, considering that data frequently include noise, the denoising autoencoder (DAE) has proven effective in enhancing performance by removing noise from observations [15]. DAEs are designed to learn robust representations of input noise [16], [17]. DAEs employ a lower-dimensional latent space than input vectors, enabling compressed data representation through efficient encoding and decoding. Their training involves intentional input corruption with noise to improve model robustness against data variability [16], [18]. In [19], a noise learning-based DAE was proposed that focuses on learning the noise in input data. This approach models the stochastic characteristics of both the original data and the noise, demonstrating significant performance enhancement.

In this paper, we consider a typical urban scenario, where multiple UAVs are deployed for source localization. These UAVs experience transitions between line-of-sight (LOS) and NLOS propagation conditions, leading to time-varying connectivity in which only a subset of UAVs maintain LOS links with the source at any given time. This dynamic channel characterization introduces realistic propagation variability into our system model. To address the challenges of data transmission in such multiple UAV networks, we propose a novel convolutional neural network (CNN)-and-residual-block-based DAE compression approach. The novelty of this approach lies in the aspect that DAE can reduce noise and compress the original observed data, which is first applied in source localization. The effectiveness of this approach is evaluated by comparing it with other approaches, including the Discrete Fourier Transform (DFT)-based approach [20], Hadamard-based approach [21] and the Principal Component Analysis (PCA)-based approach [22]. Our results demonstrate significant improvements in both compression efficiency and localization accuracy compared to existing approaches.

The remainder of this paper is organized as follows. Section II presents a system model for multiple UAV localization using the DAE-based data compression approach. Section III details the architecture of the proposed DAE network. Section IV elaborates on the simulation results to evaluate the proposed DAE approach, followed by conclusions in Section V.

Notations: Throughout this paper, we adopt the following notations. Italic letters are used to represent scalars, while bold lowercase letters are utilized for vectors, and uppercase letters for matrices. $(\cdot)^T$ denote the transpose of a vector.

II. SYSTEM MODEL

Consider a collaborative network consisting of N spatially distributed and time-synchronized UAVs² used to locate a single source. Each UAV i is in a precisely known position, denoted as $\mathbf{s}_i = (s_{ix}, s_{iy}, s_{iz})^T$, $i = 1, 2, \dots, N$, and it has knowledge of the coordinates of other UAVs. The source is located at coordinates $\mathbf{r} = (r_x, r_y, r_z)^T$, and the signals emitted by this source are received by all UAVs. However, due to the

mobility of the UAVs, the varying LOS and NLOS conditions between the UAVs and the source cause the received signals to exhibit different propagation characteristics. As a result, the received signal at the i -th UAV can be modeled as:

$$\begin{aligned} \mathbf{x}_i(t) &= \sum_{\ell=1}^{L_i} \alpha_{\ell,i} \mathbf{u}(t - \tau_{\ell,i}) + \mathbf{e}_i(t) \\ &= \mathbf{v}_i(t) + \mathbf{e}_i(t), \quad t \in [0, T] \end{aligned} \quad (1)$$

where T represents the sounding duration, L_i is the total number of specular propagation paths, $\alpha_{\ell,i}$ denotes the complex attenuation, $\tau_{\ell,i}$ denotes the delay of the ℓ -th path, $\mathbf{u}(t)$ and $\mathbf{e}_i(t)$ represent the transmitted signal and additive white Gaussian noise, respectively.

In our case, we first select UAVs operating under LOS conditions for collaborative localization. Specifically, LOS detection is performed using a multilayer perceptron for the binary classification, which effectively distinguishes UAVs with robust LOS links. Subsequently, these UAVs are utilized for TDOA calculations in source localization. In practical applications, this correlation process is often transferred to a central UAV for extraction of TDOA, imposing significant demands on the communication bandwidth. To reduce the burden on the communication link, the proposed DAE techniques are employed to compress the original sampled signals received by each UAVs³. The DAE architecture contains two components: an encoder and a decoder, which are described in detail below. Furthermore, the training procedure is also included.

Encoder: The encoder transforms the input signal $\mathbf{x}_i(t)$ into a compressed latent representation $\mathbf{z}_i(t)$, which is sequentially comprised of convolutional blocks, residual blocks, and a fully-connected output mapping. i.e., $\mathbf{h}_j = f_j(\mathbf{h}_{j-1})$ and $\mathbf{h}_0 \triangleq \mathbf{x}_i(t)$, $\mathbf{h}_J \triangleq \mathbf{z}_i(t)$. Each block implements a specific transformation $f_j(\cdot)$ for $j \in [1, 2, \dots, J]$. The initial convolutional blocks are defined as $f_j(\cdot) = \phi_j(\text{BN}_j(\mathbf{K}_j * (\cdot) + \mathbf{b}_j))$, where $\mathbf{K}_j$, $\mathbf{b}_j$, $\text{BN}_j(\cdot)$, and $\phi_j(\cdot)$ denote the convolutional kernel, bias term, batch normalization, and activation function, respectively. The subsequent residual blocks implement residual mappings defined as $f_j(\mathbf{h}_{j-1}) = \phi_j(\mathcal{F}_j(\mathbf{h}_{j-1}) + \mathbf{h}_{j-1})$, where $\mathcal{F}_j(\cdot)$ denotes the residual function. Finally, the final mapping executes a fully-connected operation: $f_J(\cdot) = \mathbf{W}_J \text{Flatten}(\cdot) + \mathbf{b}_J$, where $\mathbf{W}_J$ and $\mathbf{b}_J$ are the weight matrix and bias vector, respectively. The set of all the learnable parameters in these blocks, including all the kernels $\mathbf{K}_j$, the weight matrices $\mathbf{W}_J$, the biases $\mathbf{b}_j$, and the batch normalization parameters, is denoted by θ . Consequently, the overall encoder function $g_\theta(\cdot)$, is the composition of these individual parameterized learnable operations, i.e., $g_\theta(\cdot) = f_J \circ \dots \circ f_1(\cdot)$. This results in the final mapping:

$$\mathbf{z}_i(t) = g_\theta(\mathbf{x}_i(t)) \quad (2)$$

During the transmission process from each UAV to a central UAV, additional noise is introduced into the signal at the central UAV. The signal received by the central UAV is:

$$\mathbf{z}'_i(t) = \mathbf{z}_i(t) + \boldsymbol{\omega}_i(t) \quad (3)$$

²In practical systems, communication delays may exist, potentially causing information aging and degraded performance. However, such effects can be mitigated using techniques such as timestamping and synchronization protocols.

³The proposed DAE operates independently on each UAV, the framework can be extended to support adaptive compression strategies based on inter-UAV information exchange in future work.

where $\omega_i(t)$ represents the circularly symmetric white Gaussian noise components. In practical scenarios, due to the high signal-to-noise ratio (SNR) achievable in cooperative UAV communications and the consistent use of UAVs that employ LOS propagation for localization, it is observed that $\omega_i(t) \ll \mathbf{e}_i(t)$. Consequently, the noise introduced during the signal transfer between the UAVs and the central UAV is often neglected. Thus, we approximate $\mathbf{z}'_i(t) \approx \mathbf{z}_i(t)$ [23].

Decoder: The decoder then maps the hidden layer $\mathbf{z}_i(t)$ back to the output layer $\mathbf{y}_i(t)$ through specific functions across J layers:

$$\mathbf{y}_i(t) = g'_{\psi}(\mathbf{z}_i(t)) \quad (4)$$

In our model, the decoder is designed symmetrically with the encoder. Therefore, its details can be inferred accordingly and are omitted for brevity.

Training procedure: The purpose of DAE training is to optimize the parameters θ and ψ . This is achieved by minimizing the following loss function over the dataset:

$$\{\theta^*, \psi^*\} = \arg \min_{\theta, \psi} \frac{1}{N} \sum_{i=1}^N L(\mathbf{v}_i(t), \mathbf{y}_i(t)) \quad (5)$$

where $L(\mathbf{v}_i(t), \mathbf{y}_i(t))$ represents the reconstruction loss, which quantifies the difference between the original input $\mathbf{v}_i(t)$ and the reconstructed output $\mathbf{y}_i(t)$. Specifically, for each mini-batch, the input data is forwarded through the network to obtain the reconstructed output $\mathbf{y}_i(t)$. The loss is then calculated by comparing the real and imaginary parts of $\mathbf{y}_i(t)$ with those of the input $\mathbf{v}_i(t)$, and backpropagation is performed to update the model parameters.

III. DATA COMPRESSION BASED ON DENOISING AUTOENCODER

The DAE structure has shown promising performance in data compression and reconstruction. It typically consists of two main components: a denoising encoder, which compresses the data, and a decoder, which reconstructs it. This section provides a detailed architecture of the proposed DAE network, specifically designed for signal denoising and data compression. The architecture of the proposed CNN-and-residual-block-based DAE is illustrated in Fig. 1.

A. DAE Framework

In the encoder, the dimensions of the signals from N UAVs are reduced, and these signals are encoded into low-dimensional features denoted by M . The decoder attempts to reconstruct an output based on these features in the central UAV. The hidden layers of the network employ activation functions and complex batch normalization, while the output layer does not utilize an activation function.

The encoder employs a hierarchical structure comprising three complex-valued convolutional layers for progressive feature extraction and dimensionality reduction, two residual blocks for gradient stabilization and feature preservation, and a fully connected layer for latent space compression.

Initially, the encoder processes the original N signals, each of length $K = 1024$, using a convolutional layer with 32

channels, a kernel size of 10 and a stride of 6, which significantly reduces dimensionality while effectively capturing local patterns. Subsequently, two additional convolutional layers have 32 and 64 channels, respectively, both with a kernel size of 5 and a stride of 2. These layers further refine the features and compress the signal length. The following two blocks, similar to ResNet, consist of three convolutional layers with a residual structure, which alleviates the problem of vanishing gradients and promotes the reuse of structural features during deeper processing. The final fully connected layer maps the expanded tensor to the feature space, resulting in a feature map of dimension M . This feature map also represents the compressed data, with a compression rate (CR) of K/M , offering a substantially reduced representation compared to the original input size. The decoder is designed symmetrically with the encoder, employing deconvolutional layers to upsample the feature maps and reconstruct structural details.

B. Complexity Analysis

The computational complexity analysis of DAE is derived by calculating the complexity of each layer. Here, κ represents the size of the convolutional kernel, C_{in} and C_{out} denote the number of input and output channels of each complex convolutional layer respectively, Q represents the output length, and D represents the input length of the fully connected layer.

For the encoder architecture, the input first passes through three complex-valued convolutional layers, each with complexity of $\mathcal{O}(\kappa^{(i)} C_{\text{in}}^{(i)} C_{\text{out}}^{(i)} Q^{(i)})$, where $C_{\text{in}}^{(1)} = K$. Then it is processed by two residual blocks, each consisting of three complex convolutional layers with total complexity of $\mathcal{O}(\sum_{j=1}^3 \kappa^{(r,j)} C_{\text{in}}^{(r,j)} C_{\text{out}}^{(r,j)} Q^{(r,j)})$. Finally, the encoded feature is obtained via a fully complex-valued connected layer with a complexity of $\mathcal{O}(DM)$. Combining all components, the overall computational complexity of the encoder can be approximated as $\mathcal{O}(\sum_{i=1}^3 \kappa^{(i)} C_{\text{in}}^{(i)} C_{\text{out}}^{(i)} Q^{(i)}) + \mathcal{O}(2 \sum_{j=1}^3 \kappa^{(r,j)} C_{\text{in}}^{(r,j)} C_{\text{out}}^{(r,j)} Q^{(r,j)}) + \mathcal{O}(DM) \approx \mathcal{O}(\kappa^{(1)} K C_{\text{out}}^{(1)} Q^{(1)})$. The decoder adopts a symmetric architecture similar to that of the encoder, resulting in computational complexity equivalent to that of the encoder.

Furthermore, in our experiment, the complexity of the network was evaluated in two dimensions: the size of the model's weight parameters and the computational time processed with the same settings. All tests were performed on a server with an NVIDIA GeForce RTX 4090D GPU. The training process required approximately 30 seconds per epoch, totaling around 6000 seconds over 200 epochs. This lightweight model contains only 1.2 million trainable parameters and occupies merely 4.8 MB of storage space, making it suitable for real-world deployment in resource-constrained systems.

IV. SIMULATION RESULTS AND ANALYSIS

In this section, we first describe the experimental setup. Subsequently, we assess the localization performance of the proposed DAE based data compression approach with different CRs, and compare these results with those obtained using DFT-based, Hadamard-based, PCA-based approaches and other learning-based approaches.

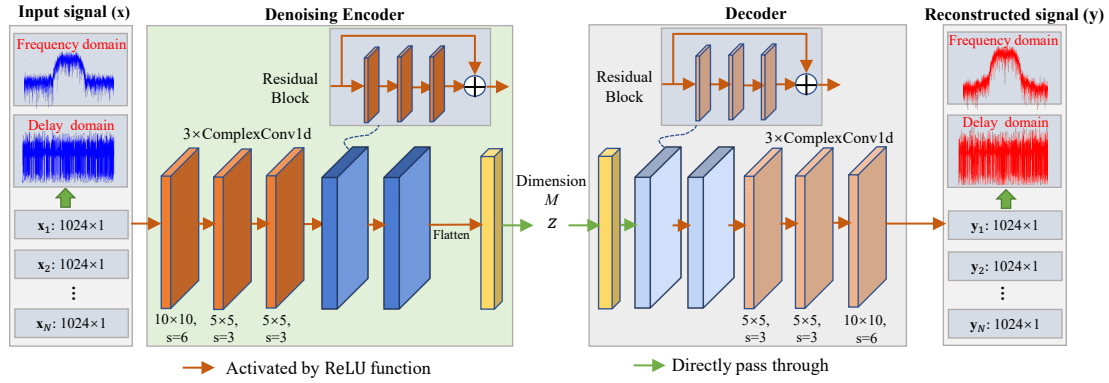


Fig. 1: The architecture of the CNN-and-residual-block-based DAE network, which includes the denoising encoder and decoder.

A. Experimental settings

The simulation was conducted in a typical urban environment using the Wireless InSite simulation platform [24]. A detailed urban crossroad scenario, aligned with the physical space, is shown in Fig. 2. The area contains 19 buildings in the scenario with an average length of 23 m and an average width of 22 m. There are 6 streets with an average length of 153 m and an average width of 22 m. The total area covers 200×260 m². We consider a scenario involving a single source and 8 UAVs for source localization. The source transmits signals using Binary Phase Shift Keying modulation with a 20 MHz bandwidth and a 2.4 GHz carrier frequency.

The configuration generated 50 snapshots of the urban crossroad scenario, each capturing the complete spatial distribution of all 8 UAVs. Throughout these snapshots, the UAVs experienced transitions between LOS and NLOS conditions, with varying numbers of UAVs maintaining LOS connections to the source at different times. This variability introduced realistic and diverse propagation conditions. To ensure statistical robustness, 200 independent Monte Carlo trials were performed for each snapshot, and each UAV collected a signal segment of 1024 samples, resulting in a total of 10000 observations. To evaluate the localization performance of the model in a wide range of SNRs, a test set was constructed with SNRs from -10 to 20 in increments of 2 dB. It is worth mentioning that the k -fold ($k=5$) cross-validation method [25] was used to optimize the model's hyperparameters, significantly improving its robustness and generalization across data subsets.

B. Selection of activation and loss functions

To optimize the performance of the DAE approach, we evaluated the impact of various activation and loss functions, including ReLU, Sigmoid, and Tanh as activation functions, and mean square error (MSE), mean absolute error (MAE), Huber loss, and cosine similarity as loss functions.

The localization performance of the proposed DAE architecture under different combinations of activation and loss functions at CR = 8 is shown in Fig. 3. The results demonstrate that the configuration using ReLU as the activation function and MSE as the loss function achieves better performance compared to other combinations. Based on this empirical analysis, ReLU and MSE were consequently selected as the preferred activation and loss functions, respectively.

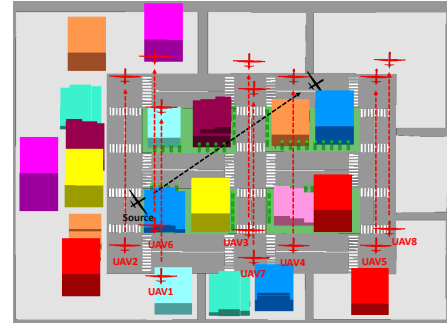


Fig. 2: Diagram of a typical urban crossroad scenario consisting numerous trees and a diverse array of buildings. 8 red UAVs and a single black source are shown, with the red and black dotted lines representing their trajectories, respectively.

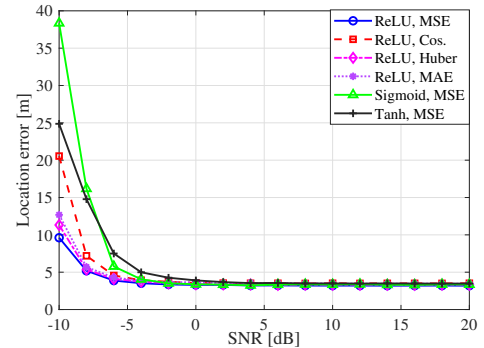


Fig. 3: Comparison of localization performance using different activation and loss functions at a CR of 8.

C. Performance evaluation

To assess the noise reduction effectiveness of the DAE, we compared the original frequency spectra of the signals with those reconstructed by the DAE. Fig. 4 shows both the original and reconstructed frequency spectra at CR = 16 using DAE for the SNR of 20 dB, 0 dB and -10 dB, respectively. It can be observed that the reconstructed signal significantly reduces noise compared to the original signal in all SNR scenarios. Furthermore, the noise reduction capability of the DAE becomes increasingly evident as the SNR decrease. Statistical evaluation of the test set under low SNR scenarios shows that the proposed DAE provides significantly equivalent SNR gain. This gain gradually decreases as the input SNR increases,

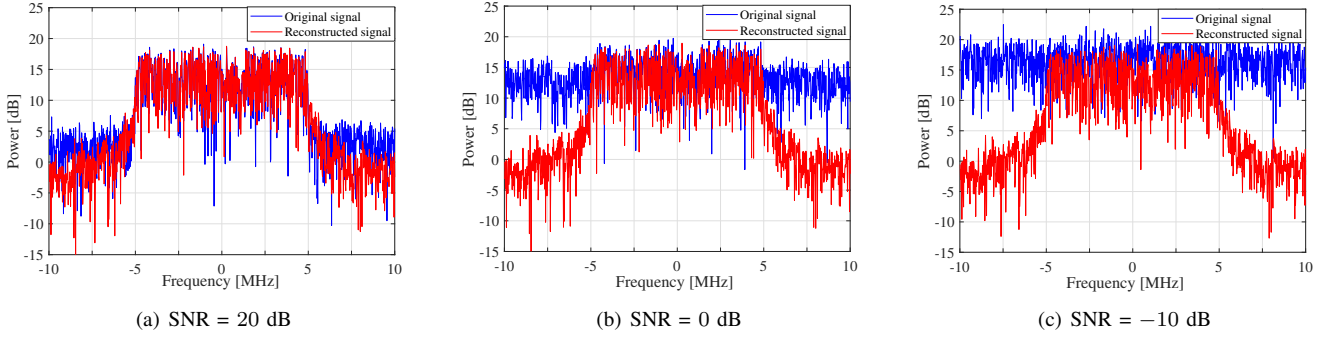


Fig. 4: Comparative analysis of frequency spectra: original signals vs. DAE reconstructed signals at a CR of 16 across SNR levels of 20 dB, 0 dB, and -10 dB, respectively.

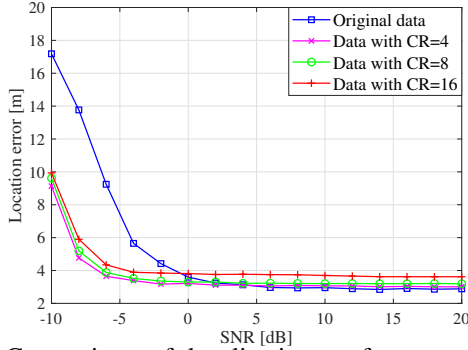


Fig. 5: Comparison of localization performance using the proposed DAE approach at different CRs.

which is consistent with the characteristics of a denoiser. The variance of the TDOA estimate is inversely proportional to the SNR [26]. An increase in SNR thus theoretically reduces the localization accuracy.

1) *Comparison of localization performance at different CRs:* Data compression reduces transmission demands and power consumption, but often at the expense of reconstruction accuracy, which can potentially reduce the effectiveness of a multiple UAV collaborative network for localization. Notably, heterogeneous computational and communication capabilities among UAVs further complicate this trade-off. For example, UAVs with limited processing resources could struggle with high CRs, reducing their effectiveness in collaborative tasks. Incorporating fairness-aware resource allocation, e.g. dynamically assigning CRs based on individual UAV capabilities, could enhance equitable participation and system robustness. Thus, collaborative systems should adapt CRs to both environmental conditions and node-specific constraints to balance positioning accuracy, compression efficiency, and fairness. Consequently, it is important to investigate the localization performance across different SNRs and CRs to optimize the trade-off between CR and localization accuracy.

Fig. 5 shows the localization performance of the proposed approach in various SNRs for different CRs of 4, 8, and 16. It is evident that higher CRs reduced localization accuracy. The localization performance achieved using the proposed DAE approach is significantly better than that obtained using the original data in low SNR scenarios, even at a CR of 16. This improvement is mainly attributed to the DAE's effective noise reduction capabilities. However, DAE exhibits slightly worse performance compared to the original data, with a negligible degradation of less than 1 m at CR=16 in high SNR scenarios.

We attribute this minor performance reduction to the inevitable loss of information during data compression. Furthermore, while the localization accuracy shows a gradual decline with increasing CR, the degradation remains limited. This demonstrates the practical applicability of the approach even at higher CRs in challenging multipath scenarios. Consequently, the DAE based approach effectively reduces communication overhead while achieving acceptable localization accuracy in high CRs, especially in low SNR scenarios.

2) *Comparison of localization performance with different data compression approaches:* To assess the performance of the proposed CNN-and-residual-block-based DAE, we compared it with DFT-based, Hadamard-based, PCA-based, deep neural network (DNN)-and-residual-block-based DAE and a CNN-based DAE approaches at a fixed CR of 16. The computational complexity of all schemes is analyzed under a consistent framework, with results summarized in Table I, where D_{hidden} denotes the length of the fully connected hidden layer. Although learning-based methods have slightly higher computational complexity than conventional DFT-based, Hadamard-based, and PCA-based approaches, the overall complexity remains within an acceptable range for practical implementation.

Table I: Comparison of computational complexity for different data compression approaches

Approach	Complexity
CNN-and-residual-block-based DAE	$\mathcal{O}(\kappa^{(1)} K C_{\text{out}}^{(1)} Q^{(1)})$
DNN-and-residual-block-based DAE	$\mathcal{O}(K D_{\text{hidden}})$
CNN-based DAE	$\mathcal{O}(\kappa^{(1)} K C_{\text{out}}^{(1)} Q^{(1)})$
DFT-based/Hadamard-based/PCA-based approach	$\mathcal{O}(KM)$

Fig. 6 presents the localization accuracy achieved by these approaches under identical conditions. The results show that both DFT-based and Hadamard-based approaches exhibit significantly degraded localization performance compared to original data, particularly in low SNR scenarios, where errors increase by 3–5 m even in high SNR scenarios. This decline is due to the substantial information loss incurred by these transforms at high CRs. Furthermore, the PCA-based approach is ineffective at high CRs. This limitation occurs because PCA relies on preserving the principal components of the multipath channel data. As the CR increases beyond a certain threshold, these critical components become compromised, leading to substantial reconstruction errors and subsequent localization

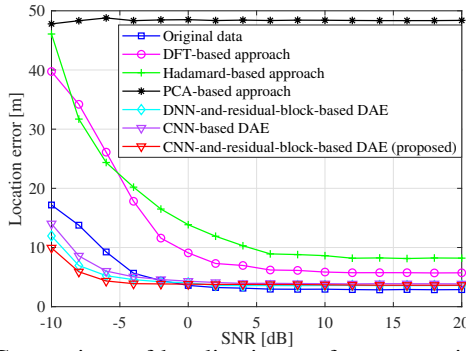


Fig. 6: Comparison of localization performance with different approaches at a CR of 16.

inaccuracies. Furthermore, we compared the proposed CNN-and-residual-block-based DAE approach with a DNN-and-residual-block-based DAE approach and a CNN-based DAE approach. The results show that the proposed approach is better than the other learning-based approaches in low SNR scenarios, while all three exhibit comparable performance in high SNR scenarios. Additionally, the CNN-and-residual-block-based DAE demonstrates significantly better localization performance than the original data in low SNR scenarios. This enhanced performance in challenging low SNR scenarios is attributed to the DAE's robust noise reduction capabilities. In high SNR scenarios, the approach still achieves localization accuracy that is highly consistent with the original data, exhibiting only slight degradation. We attribute this minor performance reduction to the inevitable loss of information during data compression.

V. CONCLUSIONS

In this study, we address the challenge of data compression for source localization in a multi-UAV network, where data are transmitted to a central UAV. The compressed data, which inherently includes noise, negatively affect the accuracy of the localization. To mitigate these issues, we propose a CNN-and-residual-block-based DAE designed to eliminate noise and reduce the dimensionality of the original data. The performance of our proposed approach is evaluated by comparing it between different CRs and against DFT-based, Hadamard-based, PCA-based approaches and other learning-based approaches. Our results demonstrate that the proposed approach significantly outperforms both uncompressed data and other approaches in low SNR scenarios, even though it employs a higher CR. Moreover, it maintains accuracy comparable to that of uncompressed data in high SNR scenarios. These results demonstrate the effectiveness of the proposed DAE for data compression in low SNR and high CR scenarios.

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